Avi Gobrin

avichaigobrin@gmail.com | linkedin.com/in/avigobrin | github.com/Avi-Gobrin | Portfolio | Canadian Citizen

Education

University of Toronto Sep 2023 - Apr 2027
Honours BSc, Mathematics & Its Applications Specialist, Statistics Major, Computer Science Minor *Toronto, ON*

Relevant Coursework: Statistical Machine Learning, Probability & Statistics, Multivariate Data Analysis, Nonlinear Optimization, Statistical Computation, Linear Algebra, Real Analysis, Mathematical Modelling, Data Structures and Algorithms

Experience

Nextdoor May 2026 - Aug 2026
Machine Learning Intern *San Francisco, CA*

- **Sole engineer** on Best Places for Families, an SEO ranking product scoring **350,000** US neighbourhoods on family-friendliness: owned the research question, timeline, data pipelines, model, evaluation, and user-facing page, and presented the system to the company.
- Designed the scoring methodology: screened 33 candidate metrics drawn from 10+ Census/ACS and internal tables down to 8 features, benchmarked OLS, ridge, NNLS, and monotone GBM, then hand-tuned the final weights when no fitted model matched human judgment.
- Built the evaluation harness for a problem with no ground truth: 100+ preregistered adversarial neighbourhood pairs resolved by pairwise Bradley-Terry judgments, cross-checked by LLM and manual review.
- Traced an early correlation between the score and wealth to a geographic join missing a county guard that had leaked out-of-state Census data into **94.6%** of neighbourhoods; the fix removed 95% of bad links and moved a housing feature by 5+ points across 44,908 neighbourhoods.
- Automated the whole path in Python: Airflow ingestion, a backend scoring service, and a Postgres/GraphQL path to a React page, run on Docker and Kubernetes on AWS with scheduled jobs, tests, and Datadog monitoring.
- Built Databricks tooling that lets the PM, design team, and non-technical reviewers observe any city's ranking and retune feature weights.

Shopify May 2025 - Aug 2025
Software Engineering Intern *Toronto, ON*

- Moved **100+** vetted merchants a day from an unassisted path into guided onboarding for Shopify's hardest case, merchants migrating an existing store from another platform, by building a full-stack partner-attach system with 2 other engineers.
- Wrote the design doc for the screening and routing: flagged large migrating merchants by subscription plan and store size, then booked each a Calendly session with a migration specialist who speaks their language, balancing their load.
- Kept Shopify's entire German merchant base compliant with new government tax forms on only **one month** of notice, shipping the program end to end in Ruby on Rails and trading polish for speed to meet the legal deadline.
- Mapped which merchant data Shopify held or could derive, built an intake form to collect the rest, and integrated the German government's tax API to fill each merchant's form and email it to them as a completed PDF.

Projects & Research

University of Toronto, Department of Statistical Sciences Aug 2026 - Present
Undergraduate Researcher, Advisor: Prof. Ricardo Baptista (Vector Institute) *Toronto, ON*

- Meet biweekly to study Transmuting Prompts into Weights (Mazzawi et al., 2026), which represents a prompt's effect as a low-rank update to a model's weights, and to develop extensions using thought vectors.
- Proposing a complement to routing among specialized models: whether the low-rank update a task induces reveals which components of a larger model it needs, so redundant computation can be pruned without losing accuracy.

Latent-Structure Modeling of Student Outcomes (OULAD) *STA437 Research Report* [Report] 2026

- Ran a full multivariate pipeline (PCA, factor analysis, Gaussian mixture clustering, LDA) over 23,301 Open University student-course registrations, recovering two dominant dimensions: assessment engagement and performance.
- Lifted minority-class recall from **3.2% to 39.3%** by benchmarking generative against discriminative classifiers under 5-fold CV (logistic regression 67.0% vs. LDA 64.5% vs. 53.0% majority baseline).

Tiny GPT From Scratch *Personal Project | Python, NumPy, pytest* [Repo] 2026

- Implemented a character-level GPT in pure NumPy with no autograd (multi-head causal attention, pre-LN blocks, Adam, top-k sampling), verifying every hand-written backward pass against finite differences in a 56-test pytest suite.

Technical Skills

Languages & Tools: Python, R, SQL, Ruby, Java, JavaScript, Bash, Git, Claude Code
ML & Statistics: PyTorch, TensorFlow, scikit-learn, statsmodels, NumPy, SciPy, pandas
Infrastructure & Web: Airflow, Databricks, Postgres, AWS, Docker, Kubernetes, Datadog, React, Ruby on Rails, GraphQL
Competitions: Putnam Mathematical Competition (Dec 2026); StarCraft II Grandmaster (top 100 in North America)